Monte Carlo DCF Enterprise Value Analysis

Oshkosh Corporation (OSK) | 50,000 simulations | Five-year forecast

All monetary values are US$ millions unless otherwise stated.

1. Executive Summary

This project applies Monte Carlo simulation to a discounted cash flow framework to estimate a probability distribution for the enterprise value of Oshkosh Corporation rather than relying on a single deterministic valuation. The simulation varies key operating and valuation assumptions across 50,000 trials and compares the resulting model-implied enterprise values with the current enterprise value of $9,060 million.

Measure	Model result	Interpretation
Mean enterprise value	$9,027.50m	0.36% below current EV
Median enterprise value	$8,986.06m	0.82% below current EV
5th percentile	$5,750.28m	Severe downside scenario
95th percentile	$12,453.60m	Strong upside scenario
Probability EV > $9,060m	48.652%	Upside probability is slightly below 50%
Probability EV ≤ $9,060m	51.348%	Downside probability is slightly above 50%

Overall conclusion. The central valuation is close to the current enterprise value, indicating that OSK appears broadly fairly valued under the simulated assumptions. The distribution is wide, however, with a 5th-to-95th percentile range of approximately $5.75 billion to $12.45 billion. This demonstrates meaningful valuation uncertainty and shows that the investment conclusion depends heavily on discount-rate and terminal-value assumptions.

Model-review warning: the current Python code treats the simulated net working capital ratio (approximately 16.6%–17.7%) as a dollar change and subtracts only about 0.17 from annual FCFF. The ratio should instead be converted into a working-capital balance and the year-on-year change should be deducted. The reported valuation distribution should therefore be treated as a preliminary model output until this formula is corrected and the simulation is rerun.

2. Project Objective

The objective is to estimate a defensible range of possible enterprise values for OSK by combining a five-year unlevered DCF with Monte Carlo simulation. The approach recognises that sales growth, profitability, capital expenditure, tax, working capital, the weighted average cost of capital, and perpetual growth are uncertain rather than fixed.

- Produce a valuation distribution instead of one point estimate.
- Measure downside, central, and upside valuation outcomes using percentiles.
- Estimate the probability that model-implied enterprise value exceeds the current enterprise value.
- Identify the assumptions that have the strongest relationship with enterprise value.

3. Historical Basis and Model Inputs

The simulation inputs were calibrated primarily from the historical financial analysis in the accompanying workbook. OSK reported 2024 sales of $10,755.5 million, operating income of $1,010.7 million, depreciation and amortisation of $200.1 million, capital expenditure of $232.0 million, and operating net working capital of $1,786.5 million. Historical analysis also showed a 2024 EBIT margin of 9.397%, capex equal to 2.157% of sales,

and operating working capital equal to 16.610% of sales. These figures provide the starting point and ranges used in the simulation.

Input	Distribution / range	Economic role	Implementation
Sales growth	Year-specific triangular ranges; modes decline from 12% to 7%	Drives forecast revenue	Independently simulated by year
EBIT margin	Triangular: 4.495% / 7.458% / 9.397%	Determines operating profit	One draw held constant across five years
WACC	Triangular: 8% / 9% / 11%	Discounts FCFF and terminal value	One draw held constant across five years
Perpetual growth	Triangular: 1.5% / 2.5% / 3.1%	Drives terminal value	One draw per simulation
Capex / sales	Triangular: 2.157% / 3.000% / 3.913%	Cash reinvestment requirement	One draw held constant across five years
Tax rate	Triangular: 21.0% / 23.3% / 26.0%	Converts EBIT to NOPAT	One draw held constant across five years
Operating NWC / sales	Triangular: 16.6% / 16.7% / 17.7%	Working-capital investment	Requires formula correction
D&A	$200.1m grown with cumulative sales	Non-cash add-back	Scales with revenue growth

4. Methodology

For each of 50,000 trials, the model generates a five-year financial path and calculates enterprise value using the following sequence:

- Forecast sales by compounding the simulated annual sales-growth rates from the 2024 revenue base.
- Estimate EBIT as forecast sales multiplied by the simulated EBIT margin.
- Calculate NOPAT as EBIT × (1 − tax rate).
- Estimate FCFF as NOPAT + depreciation and amortisation − capital expenditure − change in net working capital.
- Discount each annual FCFF by the simulated WACC.
- Calculate terminal value using the Gordon growth model: Terminal FCFF ÷ (WACC − perpetual growth).
- Add the present value of forecast-period FCFF and terminal value to obtain model-implied enterprise value.

Core equation: Enterprise Value = $\Sigma[FCFF_t \div (1 + WACC)^t] + [\text{Terminal Value} \div (1 + WACC)^5]$.

5. Simulation Results

The simulated distribution is approximately bell-shaped around the central valuation, with a visible right tail. The median of $8,986.06 million is slightly lower than the mean of $9,027.50 million, consistent with modest positive skew from high-value terminal scenarios. The histogram marks the 5th, 50th, and 95th percentiles at $5,750.28 million, $8,986.06 million, and $12,453.60 million, respectively.

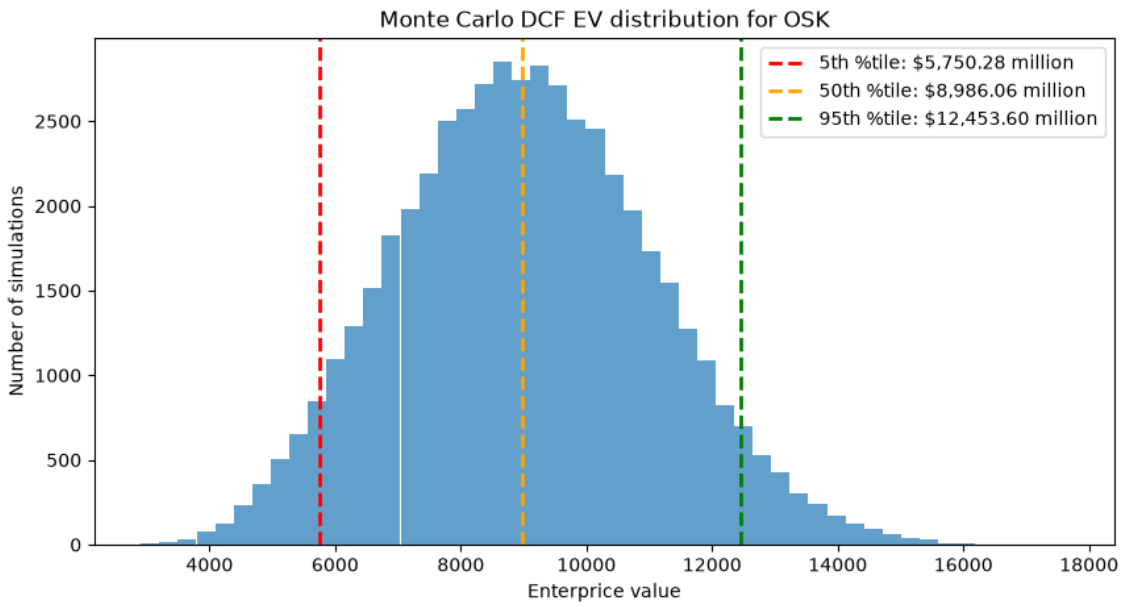


Figure 1. Monte Carlo DCF enterprise-value distribution for OSK. The red, orange, and green dashed lines represent the 5th, 50th, and 95th percentiles.

Scenario interpretation

Scenario definition	Enterprise-value range	Interpretation
Lower third / conservative	Below $8,089.05m	Weak operating or valuation assumptions; below current EV
Middle third / base	$8,089.05m–$9,890.45m	Central range spanning the current EV of $9,060m
Upper third / optimistic	Above $9,890.45m	Favourable operating performance and/or valuation assumptions
90% probability interval	$5,750.28m–$12,453.60m	Broad uncertainty range between 5th and 95th percentiles

Market comparison. The current enterprise value of $9,060 million is $32.50 million above the simulation mean and $73.94 million above the median. The probability of model-implied enterprise value exceeding the current value is 48.652%, while the probability of being equal to or below the current value is 51.348%. The model therefore does not indicate a clear valuation advantage at the current EV; instead, it suggests a nearly balanced risk-reward profile.

6. Sensitivity Analysis

The reported correlation matrix indicates that WACC has the strongest measured relationship with enterprise value, with a correlation of approximately −0.399. This is economically intuitive: a higher discount rate reduces both the present value of forecast cash flows and the terminal value. The individual annual sales-growth correlations shown in the output are much smaller, with Year 5 sales growth at approximately +0.033.

The printed sensitivity matrix is incomplete. It excludes EBIT margin, capex percentage, tax rate, working-capital assumption, and depreciation, even though these variables affect FCFF. Perpetual growth appears close to zero in the displayed correlation output, which is inconsistent with standard DCF mechanics and should be checked by printing the full Enterprise Value correlation row and validating data alignment. For a stronger project, use rank correlation, standardized regression, or a tornado chart with all simulated inputs.

7. Investment Interpretation

- Central valuation: Mean and median enterprise values are close to the current EV, suggesting that OSK is approximately fairly valued under the modelled assumptions.
- Downside risk: The 5th-percentile EV of $5.75 billion is about 36.5% below the current EV, showing material downside under adverse combinations of operating performance and valuation assumptions.
- Upside potential: The 95th-percentile EV of $12.45 billion is about 37.5% above the current EV, indicating meaningful upside if growth, profitability, and terminal assumptions are favourable.
- Decision quality: Because the probability of exceeding current EV is only 48.652%, the simulation does not provide a strong standalone buy signal.
- Key value driver: WACC is the strongest measured driver, highlighting the importance of capital-market conditions and discount-rate selection.
- Use in practice: The output is most useful as a valuation-risk framework for investment screening, negotiation ranges, scenario planning, and identifying assumptions requiring further due diligence.

8. Model Limitations and Required Corrections

Issue	Recommended treatment
Working-capital formula	The model simulates an NWC-to-sales ratio but subtracts the ratio itself from FCFF. Forecast NWC should equal forecast sales × NWC ratio, and change in NWC should equal $NWC_t - NWC_{t-1}$.
Input dependence	The model largely treats inputs as independent. In reality, sales growth, margins, capex, working capital, WACC, and terminal growth may be correlated.
Constant annual parameters	EBIT margin, capex ratio, tax rate, and WACC are each held constant across all five forecast years within a trial, which may understate path-specific operating changes.
Terminal-value concentration	A large proportion of DCF enterprise value is typically generated by terminal value, making the result highly sensitive to WACC and perpetual growth.
Sensitivity coverage	The current sensitivity DataFrame omits several major inputs and reports simple Pearson correlations only.
Reproducibility	No random seed is set. Results will change slightly each time the model is run.
Distribution calibration	Triangular ranges are based mainly on a short historical period and judgement; these ranges should be stress-tested and documented.
Enterprise versus equity value	The model estimates enterprise value. Net debt and other claims must be deducted before deriving equity value or value per share.

9. Recommended Next Steps

- Correct the working-capital calculation and rerun all 50,000 simulations.
- Set a random seed, for example np.random.seed(42), so reported results are reproducible.
- Add EBIT margin, capex, tax, NWC, and D&A assumptions to the sensitivity dataset.
- Report the standard deviation and additional percentiles, including the 25th and 75th percentiles.
- Calculate the terminal-value share of total enterprise value for each simulation and flag extreme cases.

- Introduce realistic correlations among operating assumptions, especially growth, margins, and reinvestment.
- Convert enterprise value to equity value and implied value per share by deducting net debt and other claims.
- Compare the Monte Carlo DCF range with comparable-company valuation to triangulate the final conclusion.

10. Conclusion

The Monte Carlo DCF provides a more informative valuation framework than a single-case DCF because it quantifies both the central estimate and the dispersion of possible outcomes. Based on the reported run, OSK has a mean model-implied enterprise value of $9.03 billion and a median of $8.99 billion, both close to the current EV of $9.06 billion. The 48.652% probability of exceeding current EV suggests no decisive undervaluation signal, while the wide 5th-to-95th percentile range demonstrates substantial uncertainty. The preliminary investment conclusion is therefore neutral: OSK may offer upside under favourable assumptions, but the valuation does not provide a clear margin of safety. This conclusion should be revisited after correcting the working-capital treatment and expanding the sensitivity analysis.

Appendix: Correct Working-Capital Logic

The intended treatment is to forecast the operating NWC balance as a percentage of sales and deduct only the annual increase in that balance from FCFF:

$$\textit{Forecast } NWC_t = \textit{Forecast } Sales_t \times \textit{NWC-to-Sales Ratio}$$
$$\textit{Change in } NWC_1 = \textit{Forecast } NWC_1 - \textit{Initial NWC}$$
$$\textit{Change in } NWC_t = \textit{Forecast } NWC_t - \textit{Forecast } NWC_{t-1}$$

Using initial operating NWC of approximately $1,786.5 million from the workbook would align the forecast with the historical NWC-to-sales ratio and prevent FCFF from being overstated.